

Lecture - 4

CO4: Compare various statistical methods of study data samples

CO5: Analyze and evaluation of different sets of data using hypothesis testing.

JOINT, AND MARGINAL DISTRIBUTION FUNCTION FOR DISCRETE RANDOM VARIABLE

Two-Dimensional Joint Distribution Function

In analogy with the distribution function $F(x) = P[X \leq x]$ of one-dimensional random variable X already discussed, the distribution function of the two-dimensional random variable (X, Y) for all real x and y is defined as

$$F(x, y) = P[X \leq x, Y \leq y]$$

Marginal Distribution Functions

Let (X, Y) be a two-dimensional discrete random variable having $F(x, y)$ as its distribution function. Now **the marginal distribution function of X** is defined as

$$F(x) = P[X \leq x]$$

$$F(x) = P[(X \leq x, Y = y_1)] + P[(X \leq x, Y = y_2)] + \dots$$

$$F(x) = \sum_j P[X \leq x, Y = y_j]$$

Similarly, the **marginal distribution function of Y** is defined as

$$F(y) = P[Y \leq y]$$

$$F(x) = P[(X = x_i, Y \leq y)] + P[(X = x_2, Y \leq y)] + \dots$$

$$F(x) = \sum_i P[X = x_i, Y \leq y]$$

Example 1: Considering the probability distribution function

	Y	1	2
X			
1		0.1	0.2
2		0.1	0.3
3		0.2	0.1

find i) $F(2, 2)$, $F(3, 2)$ ii) $F_X(3)$ iii) $F_Y(1)$

Solution:

$$\text{i) } F(2, 2) = P[X \leq 2, Y \leq 2]$$

$$F(2, 2) = P[X = 2, Y \leq 2] + P[X = 1, Y \leq 2]$$

$$F(2, 2) = P[X = 2, Y = 2] + P[X = 2, Y = 1] + P[X = 1, Y = 2] + P[X = 1, Y = 1]$$

$$F(2, 2) = 0.3 + 0.1 + 0.2 + 0.1 = 0.7$$

$$F(3, 2) = P[X \leq 3, Y \leq 2]$$

$$F(3, 2) = P[X \leq 2, Y \leq 2] + P[X = 3, Y \leq 2]$$

= 0.7 + P[X = 3, Y ≤ 2] [∵ first term on R.H.S. has been obtained in part (i) of this example]

$$= 0.7 + P[X = 3, Y = 2] + P[X = 3, Y = 1] = 0.7 + 0.1 + 0.2 = 1$$

$$\text{ii) } F_X(3) = P[X \leq 3] = P[X \leq 3, Y = 1] + P[X \leq 3, Y = 2]$$

$$F_X(3) = P[X = 3, Y = 1] + P[X = 2, Y = 1] + P[X = 1, Y = 1] + P[X = 3, Y = 2] \\ + P[X = 2, Y = 2] + P[X = 1, Y = 2]$$

$$= 0.2 + 0.1 + 0.1 + 0.1 + 0.3 + 0.2 = 1$$

$$\text{ii) } F_Y(1) = P[Y \leq 1] = P[X = 1, Y \leq 1] + P[X = 2, Y \leq 1] + P[X = 3, Y \leq 1]$$

$$F_Y(1) = P[X = 1, Y = 1] + P[X = 2, Y = 1] + P[X = 3, Y = 1]$$

$$= 0.1 + 0.1 + 0.2 = 0.4$$

Example 4: Find the joint and marginal distribution functions for the joint probability distribution.

$P[X = 0, Y = 0] = 2/9$, $P[X = 0, Y = 1] = 1/9$, $P[X = 1, Y = 0] = 1/9$, and $P[X = 1, Y = 1] = 5/9$.

Solution: For the joint distribution function, we have to find

$F(x, y) = P[X \leq x, Y \leq y]$ for each x and y , i.e. we are to find $F(0, 0)$, $F(0, 1)$, $F(1, 0)$, $F(1, 1)$.

$$F(0, 0) = P[X \leq 0, Y \leq 0] = P[X = 0, Y = 0] = \frac{2}{9}$$

$$\begin{aligned} F(0, 1) &= P[X \leq 0, Y \leq 1] = P[X = 0, Y = 0] + P[X = 0, Y = 1] \\ &= \frac{2}{9} + \frac{1}{9} = \frac{3}{9} \end{aligned}$$

$$\begin{aligned} F(1, 0) &= P[X \leq 1, Y \leq 0] = P[X = 1, Y = 0] + P[X = 0, Y = 0] \\ &= \frac{1}{9} + \frac{2}{9} = \frac{3}{9} \end{aligned}$$

$$\begin{aligned} F(1, 0) &= P[X \leq 1, Y \leq 0] = P[X = 1, Y = 0] + P[X = 0, Y = 0] \\ &= \frac{1}{9} + \frac{2}{9} = \frac{3}{9} \end{aligned}$$

$$\begin{aligned} F(1, 1) &= P[X \leq 1, Y \leq 1] \\ &= P[X = 1, Y = 1] + P[X = 1, Y = 0] + P[X = 0, Y = 1] \\ &\quad + P[X = 0, Y = 0] \\ &= \frac{5}{9} + \frac{1}{9} + \frac{1}{9} + \frac{2}{9} = 1 \end{aligned}$$

Above distribution function $F(x, y)$ can be shown in the tabular form as follows:

	$Y \leq 0$	$Y \leq 1$
$X \leq 0$	$\frac{2}{9}$	$\frac{3}{9}$
$X \leq 1$	$\frac{3}{9}$	1

Marginal distribution function of X is obtained on finding $F(x) = P[X \leq x]$ for each x , i.e. we have to obtain $F_X(0), F_X(1)$.

$$\text{ii) } F_X(0) = P[X \leq 0] = P[X = 0]$$

$$= P[X = 0, Y = 0] + P[X = 0, Y = 1] = \frac{2}{9} + \frac{1}{9} = \frac{3}{9}$$

$$F_X(1) = P[X \leq 1] = P[X \leq 1, Y = 0] + P[X \leq 1, Y = 1]$$

$$= P[X = 1, Y = 0] + P[X = 0, Y = 0] + P[X = 1, Y = 1] + P[X = 0, Y = 1]$$

$$= \frac{1}{9} + \frac{2}{9} + \frac{5}{9} + \frac{1}{9} = 1$$

$\therefore$ marginal distribution function of X is given as

X	$F(x)$
≤ 0	$3/9$
≤ 1	1

Similarly, marginal distribution function of Y can be obtained.